

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Safety Studies and Monsoon Combat Plan

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Rev No.	Date	Description of Revision
R01	10.08.22	Issued for Review
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Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-PLN-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

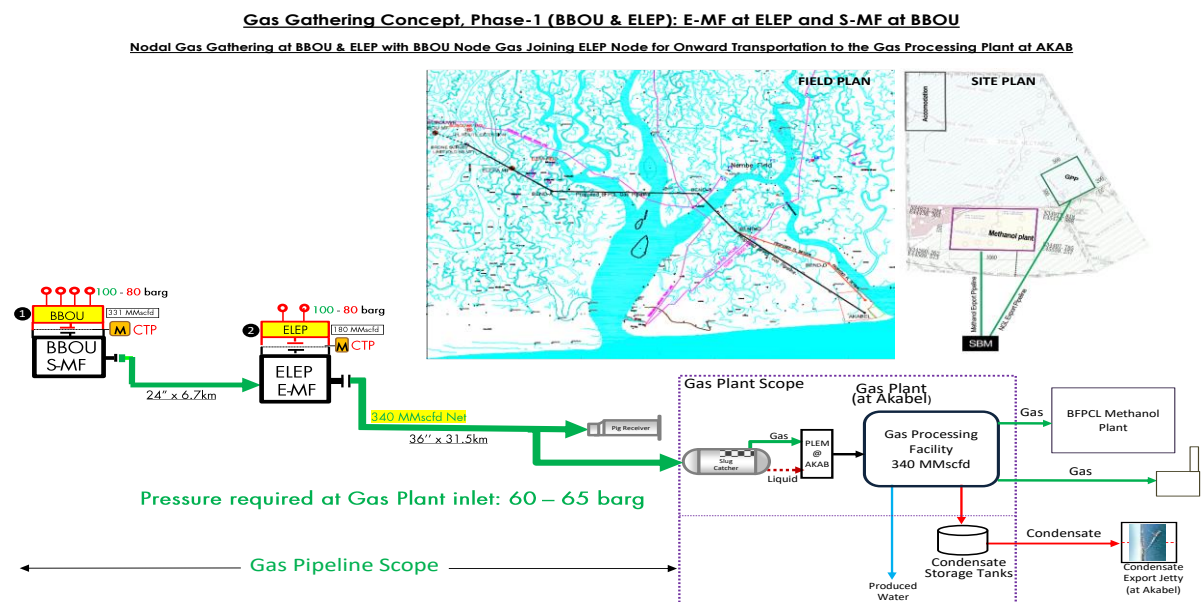


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to highlight the safety studies carried out as part of this FEED work, and to explain how the risk of flooding has been addressed in the FEED.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Safety Studies

In addition to the safety study documented in this report and addressing the project mitigation measures against heavy rain/flooding, the following additional Safety studies have been carried out and appropriately documented during the FEED.

The reports are available in WRENCH for reference.

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-THS-PRC-001	Fire and Explosion Risk Assessment
EA-BFPCL-GPMC-GGPL-FEED-THS-LST-001	Hazardous Area Classification Schedule
EA-BFPCL-GPMC-GGPL-FEED-THS-DWG-001	Hazardous Area Classification Drawing -BBOU, ELEP, AKAB
EA-BFPCL-GPMC-GGPL-FEED-THS-DWG-002	Safety Equipment & Escape Route Layout - BBOU, ELEP, AKAB
EA-BFPCL-GPMC-GGPL-FEED-THS-PLN-003	Emergency Response / Disaster Management Plan
EA-BFPCL-GPMC-GGPL-FEED-HSE-PLN-003	Effluent & Emission Requirement & Compliance Plan
EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-012	HAZID Terms of Reference and HAZID Review
EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-002	Project Hazard & Effect Register (HAZID Report)
EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-011	HAZOP Terms of Reference and HAZOP Review
EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-004	HAZOP Report
EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-005	SIL Classification
EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-006	Design ALARP Demonstration (HSE Case)

4. Monsoon Combat Plan

4.1. Wet (Rainy) Season

The Wet season (also known as rainy season) which occurs between the months of March and October every year is the equivalent of Monsoon season in Nigeria.

In the Niger Delta area of Nigeria, light rainfall begins in March and increases in intensity through June and July. A brief break is experienced in August, to begin again in September and the season doesn't end until late October.

Rainy season in Nigeria has its advantages as well as its challenges. Some of the advantages, or benefits, of the rainy season in Nigeria include:

- It helps in both crop and animal (land and aquatic) farming.
- It gives relief to people from the hot, or dry, season.
- There is more water to power the hydroelectric power generating plants, amongst others.

The challenges of the rainy season include:

- Flooding. Flooding has become a major issue in Nigeria as many a time, flooding claims human lives and destroys properties.
- Upsurge in malaria and other sickness. There is an increase in malaria sickness during the rainy season as mosquitoes breed on stagnant water which is in abundance during the rainy season.
- Ecological disasters like soil erosion and collapse of trees are brought about by heavy rains.

Of the challenges enumerated above, the one that affects the project the most, and in particular the facilities, is Flooding.

The worst flooding recorded in the project area (Niger Delta area) in recent times occurred in 2012 with unprecedented water level approaching 5 metres recorded in the area. Many facilities were submerged in water and sensitive equipment and items destroyed.

4.2. Flood Management

Flooding is usually caused by, or affects, much wider areas of communities as the different streets and waterways are interconnected and feed into each other. Flooding is therefore best tackled at town level to ensure no section of the town poses a risk to the other.

Some measures could be proactively taken by local and national authorities in order to combat the dangerous effect of flooding. Some of the measures include (i) proper town planning, (ii) provision of drainage channels, (iii) building of dams, (iv) Wing Dykes, (v) Diversion Spillways, (vi) Afforestation, and (vii) Artificial Levee.

Proper town planning ensures orderly development of a town and helps avoid indiscriminate erection of structures which block drainage channels.

Drainage channels help in the removal of surface water from an area with excess water.

Dams are the classic hard engineering solution to flooding problems. A dam is a giant wall built across a river's channel to impede its flow.

Wing Dykes are slats that are placed in a river's channel at 90 to the banks. They help reduce the risk of flooding by getting water away from an area at risk of flooding as quickly as possible, preventing a build-up of water.

Diversion spillways are artificial channels that a river can flow into when its discharge rises.

Afforestation involves the planting of trees in drainage basins to increase interception and storage while reducing surface runoff.

A levee is a low wall built at the side of a river to prevent it from flooding.

4.3. Impact Mitigation

A private Company may not be able to do much to combat flooding in an area as the causes of flooding are not restricted to its immediate locality, but it can take measures to protect its facilities from the impact of flooding. Some of the measures such a company can take include proper layout design to ensure adequate drainage within its property limits, and proper elevation of the facilities relative to the general elevation of the surrounding infrastructure.

For the Gas Gathering Pipelines and Associated Infrastructure, the terrain is swamp and the facilities will be located on a piled steel platform covered with steel gratings that are open to the river directly below. There is therefore no possibility of water retention on the platform and hence layout design, slopes, etc. is not much of a consideration with respect to flooding.

Given the sequence of activities that led to its occurrence, and the measures subsequently put in place by the relevant authorities to combat it, the 2012 flooding is taken as a one-off occurrence which is not likely to be repeated. This fact was recognised and accounted for during the HAZID (Hazard Identification) study carried out as part of the FEED.

To this effect, the water level attained during that period has been taken as the maximum possible water level for the project, and the manifold platforms in BBOU and ELEP have been elevated slightly above that level. This will ensure that the platforms and the equipment on the platforms always remain above water and will never get submerged and damaged during the rainy season.

Equipment and items that are susceptible to damage or reduced performance during heavy rains will either be made weatherproof or provided with shelters to keep out the rains.

4.4. Operations During Flooding

The manifold platforms have been elevated above the highest expected water level, and hence safe from flooding but access to the facilities may prove a challenge as transportation from the main operational base in Akabel to the remote manifolds may not be feasible or safe. In that case, a shutdown of the field operations is recommended until the floods recede.